

Hybrid RAG System - Enhanced Evaluation Report

Generated: 2026-02-02 22:26:09

Total Questions Evaluated: 100

 NOVEL = Innovative metrics introduced in this evaluation

Overall Performance Summary

Standard Metrics



Novel Metrics



LLM-as-Judge Evaluation

An LLM evaluator rates each answer on 4 quality dimensions (1-5 scale).

Dimension	Avg Score	Description
Accuracy	0.00/5	Factual correctness of the answer
Completeness	0.00/5	Coverage of all relevant aspects
Relevance	0.00/5	Direct relevance to the question
Coherence	0.00/5	Logical flow and clarity

Overall LLM-as-Judge Score: 0.00/5

Adversarial Testing Results

Tests system robustness against paraphrased, negated, and unanswerable questions.

Test Type	Score	Interpretation
Paraphrase Robustness	20.0%	Needs Improvement
Negation Handling	10.0%	Needs Improvement

Hallucination Risk	0.0%	Low Risk (Good)
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Overall Robustness Score: 43.3%

 Novel Metric Justifications

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4. Entity Coverage Ratio

Why Novel: Unlike standard metrics, Entity Coverage focuses on NAMED ENTITIES (people, dates, numbers, monetary values) which are critical for factual accuracy in Wikipedia-based Q&A.

Calculation: Extracts entities from context and answer using regex patterns, then computes: $|Context\ Entities \cap Answer\ Entities| / |Context\ Entities|$

Advantages over ROUGE-L: Focuses on high-value factual content rather than all n-grams. A paraphrased answer with correct entities scores high.

5. Answer Groundedness Score

Why Novel: Measures whether the generated answer is semantically derived from the retrieved context, detecting potential hallucinations.

Calculation: Cosine similarity between answer embedding and CONTEXT embedding (not reference answer).

Advantages: Unlike Semantic Similarity (answer vs reference), this catches hallucinations where the model generates plausible but unsupported content.

6. LLM-as-Judge

Why Novel: Automated evaluation using an LLM to judge answer quality across multiple dimensions, more nuanced than single-score metrics.

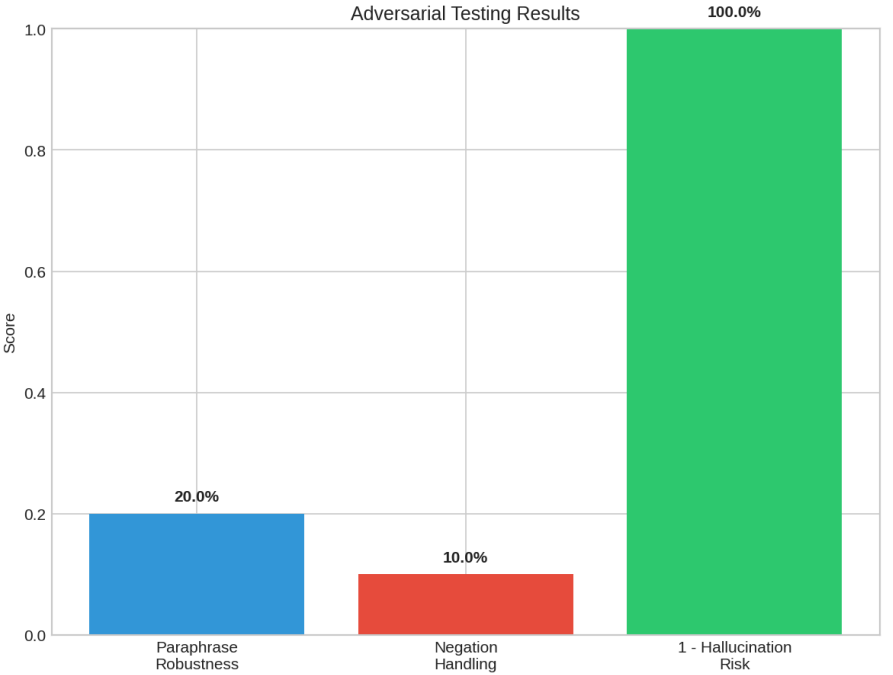
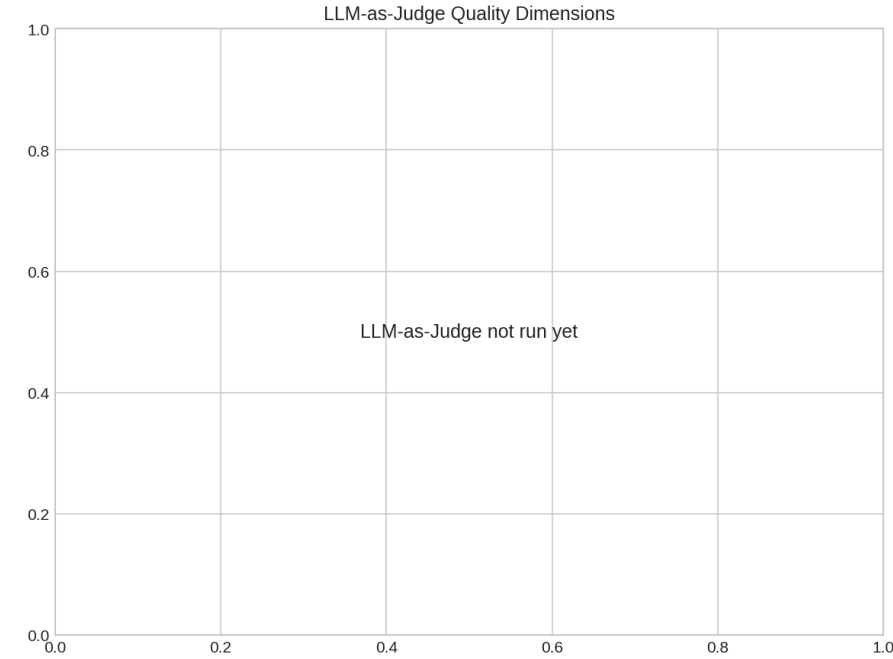
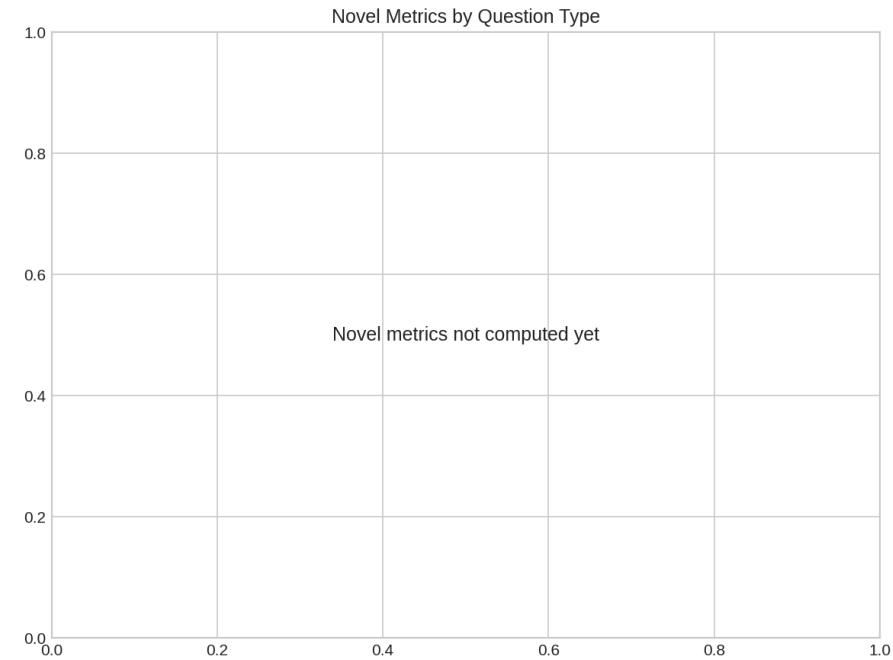
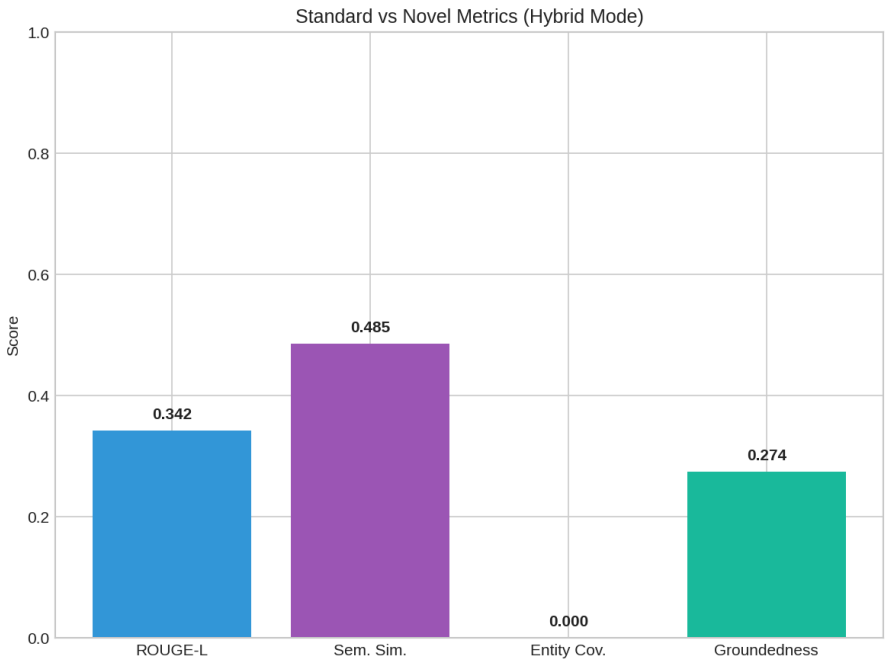
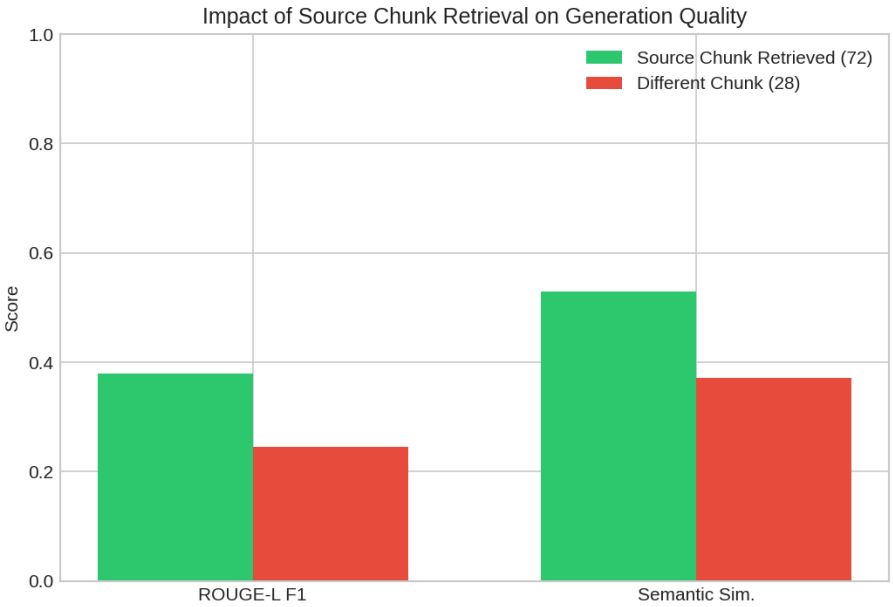
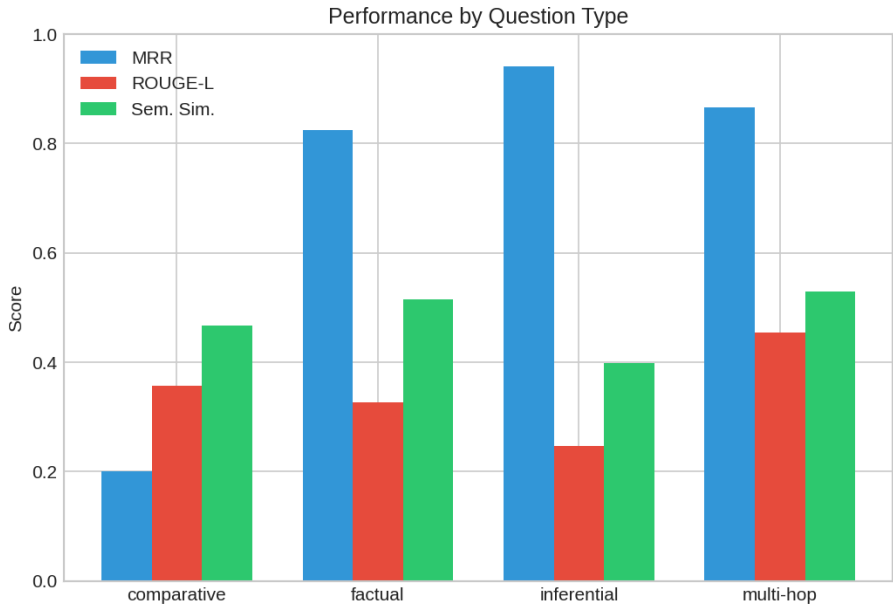
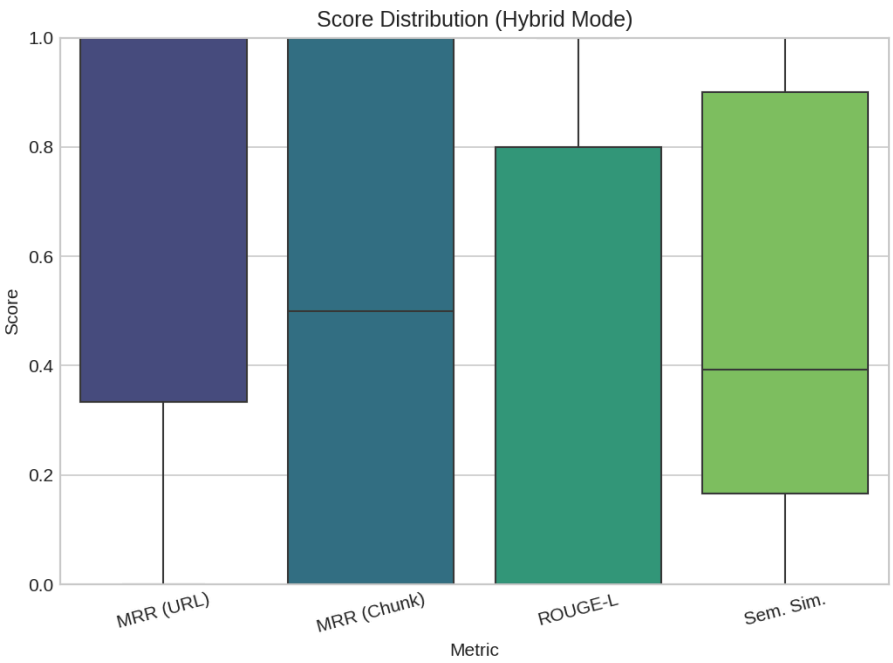
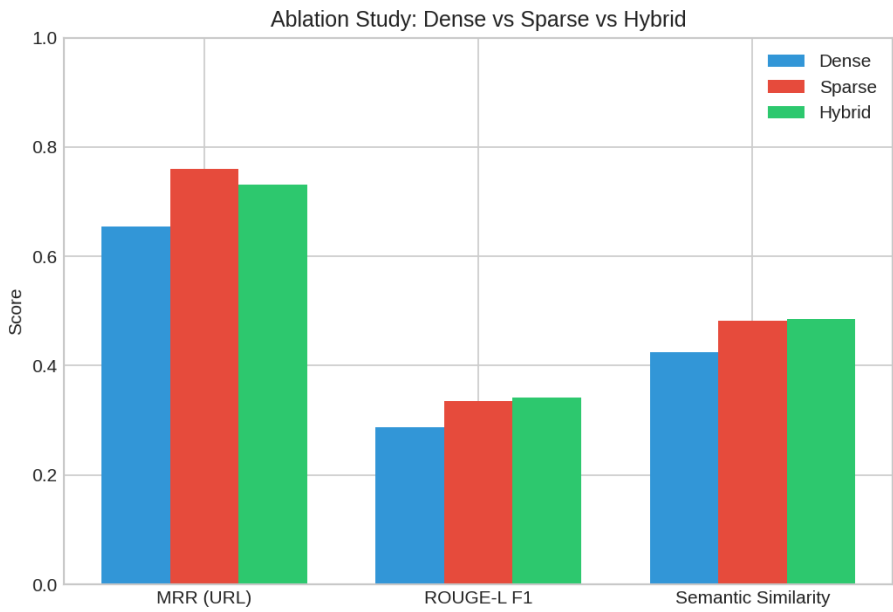
Dimensions: Accuracy, Completeness, Relevance, Coherence - each rated 1-5.

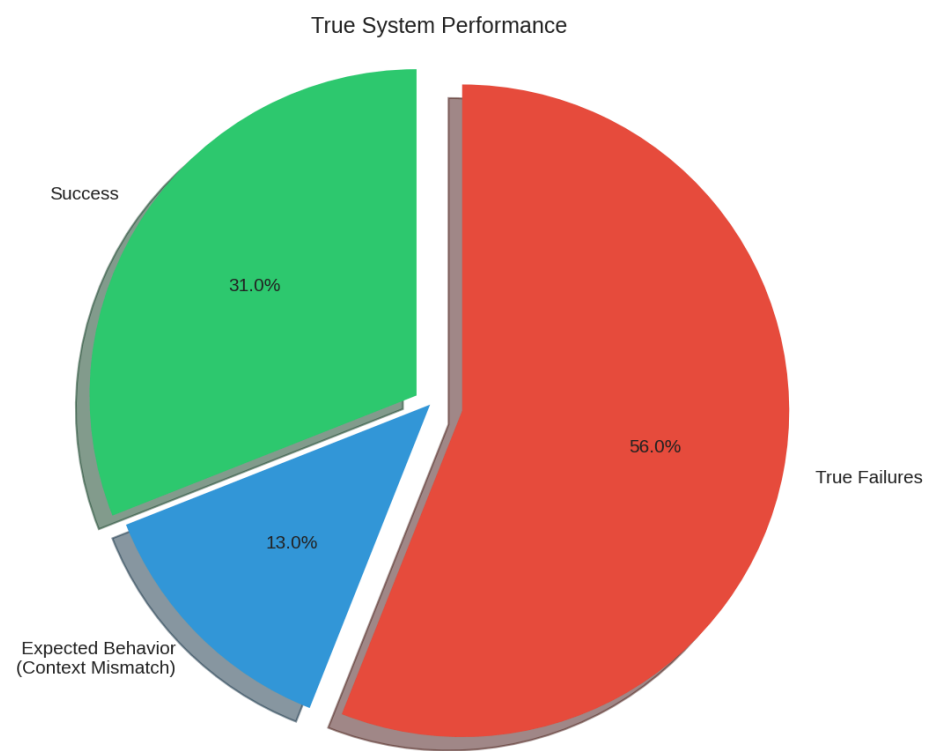
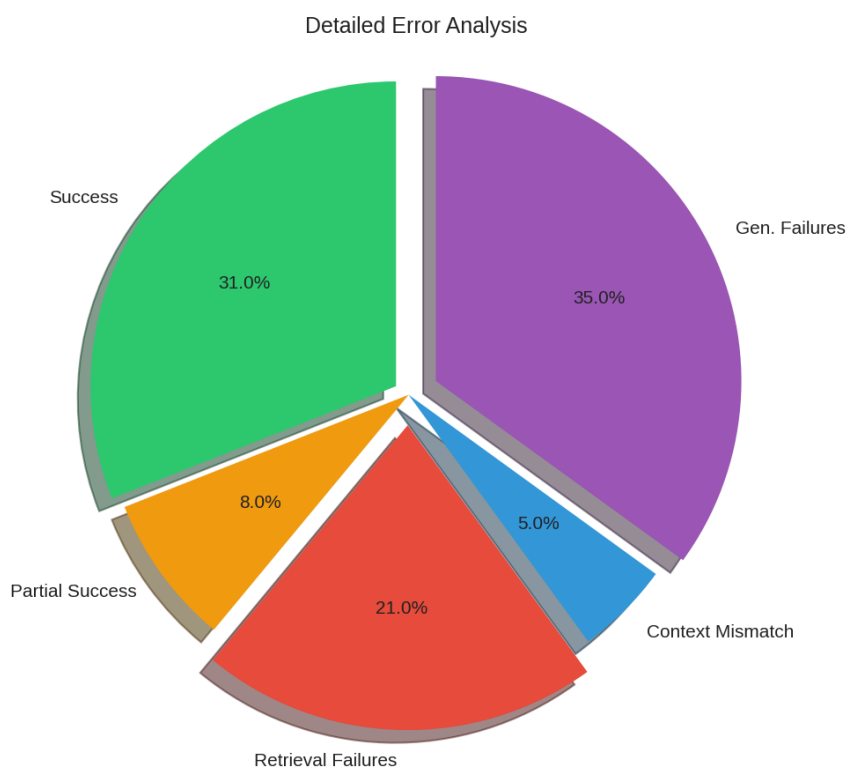
Advantages: Captures aspects of quality that are difficult to measure with n-gram or embedding-based metrics alone.

 Ablation Study: Dense vs Sparse vs Hybrid

Mode	MRR (URL)	ROUGE-L F1	Semantic Similarity	Avg Time (s)
Dense	0.6550	0.2867	0.4250	0.284
Sparse	0.7600	0.3358	0.4818	0.261
Hybrid	0.7317	0.3422	0.4850	0.278

 Visualizations





Hybrid RAG System - BITS Pilani Conversational AI Assignment 2

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metrics and evaluations designed for this assignment